

Jiwon Hwang

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SECURITY ENGINEER & SYSTEM ADMINISTRATOR

Provide technical leadership in security and system administration for enterprise operations, offering expertise ranging from cloud and AI threat analysis to binary exploitation, vulnerability assessments, and audit readiness. Identify weaknesses and select solutions to strengthen controls and mitigate risk across large infrastructure environments. Collaborate directly with C-suite, engineers, and external stakeholders to facilitate critical initiatives such as corporate spin-offs, certifications, or compliance audits.

Expertise: Enterprise-Scale Infrastructure Management ▪ Security Engineering ▪ Cloud Architecture ▪ Security Testing
Vulnerability Assessments ▪ Remediation Planning ▪ Problem-Solving ▪ Process Improvements ▪ Workflow Optimization
AI Architecture & Security ▪ Automation ▪ Cross-Team Collaboration ▪ Communications ▪ Auditing & Compliance

Select Highlights

- **Operated a 200+-node Linux HPC environment that securely supported proprietary genomics research**, including managing NVIDIA GPU clusters, petabyte-scale storage, and 100G InfiniBand networking.
- **Led Cloud security and web application testing** across AWS attack-chain scenarios, authentication flows, injection vulnerabilities, and privilege escalation paths to uncover gaps and recommend remediation solutions.
- Built defensive controls for Entra ID, Intune, centralized logging, conditional access, and audit compliance programs in order to **achieve zero findings during 4 consecutive audit cycles**.
- **Supported production AI workloads on NVIDIA GPU infrastructure**, identifying solutions to address CUDA dependency issues, scheduler failures, and resource contention in multi-user environments.
- **Conducted research at University of Maryland focused on LLM-integrated security analysis**, transformer fine-tuning, evaluation framework design, and security failure modes in agentic systems.

Professional Experience

UNIVERSITY OF MARYLAND, COLLEGE PARK – College Park, MD

2024-2026

Research Assistant, SEED Lab / Graduate Student

- Worked under Professor Yonghwi Kwon, investigating Kubernetes configuration drift where GitOps-based reconciliation overrode emergency security patches, creating gaps between actual system state and audit reports.
- Utilized automated hourly GitHub action pipelines to validate performance across multiple scenarios, highlighting operational risks in production environments.

THERAGENBIO – Pangyo, South Korea

2020-2024

System Administrator

- Provided infrastructure management across 200+ servers, overseeing rack installation, firmware updates, and refresh cycles for a petabyte-scale storage environment and 100G InfiniBand network supporting ~120 employees.
- Defined and implemented companywide strategy to transition cloud security to M365 and Azure AD. Balanced security requirements with usability across MDM, MAM, and conditional access processes and protocols.
- Directed a cloud pilot program to validate new policies and launched an Azure AD log monitoring pipeline to track end-user activity and detect suspicious behavior post-deployment.
- Planned and led 100+ server migrations with zero data loss to support a corporate spin-off. Documented operational workflows, coordinated downtime, and reconfigured network stack for new infrastructure setup.
- Developed 4K+-line modular Python/Bash automation framework supporting the full server onboarding process; cut provisioning time from 4 hours to 30 minutes while eliminating duplicate documentation efforts.
- Achieved zero audit failures while leading ISO 27001, ISO 27701, and GCLP audits across 4 consecutive cycles.
- Architected an HTML replica login page that captured 85% of employee credentials for a phishing simulation program. Developed email reporting mechanism and security awareness training based on program results.
- Led distributed storage efficiency tests to determine optimal small-file handling configurations for genomics workloads.

KOREA INFORMATION SECURITY MANAGEMENT INSTITUTE (KISMI) – Seoul, South Korea

2020

Security Audit & Penetration Testing Consultant

- Managed the ISMS compliance cycle for SK Telecom, including conducting internal audits, vulnerability assessments, and policy development across the enterprise telecommunication infrastructure.
- Led an ethical hacking project for InBody, identifying and documenting web application vulnerabilities by conducting SQL injection, XSS, authentication bypass, and parameter manipulation.
- Performed a compliance assessment for sustained certification validity at KAKAO VX, including leading web and app penetration testing to uncover vulnerabilities across HTTP/HTTPS services.
- Incorporated ISO 27017/27018 cloud security standards into company policies and created audit-related documentation.

Research & Special Projects

- **AWS Cloud Security Assessment** (Spring 2026): Completed a full-scope assessment of the AWS e-commerce platform. Identified 56 vulnerabilities across data security, VM analysis, network segmentation, and disaster recovery planning. Generated a 91-page report with remediation recommendations mapped to PCI-DSS, GDPR, and NIST 800-53.
- **Binary Exploitation** (Fall 2025): Implemented buffer overflow, format string, GOT/PLT hijacking, and heap exploitation in C to generate reliable shell across all scenarios. Debugged and documented exploits in LaTeX lab reports.
- **CTF Penetration Test** (Fall 2025): Performed black-box pen test against a Linux target, including SQL injection, brute force, web shell, password spraying, and hash cracking. Captured 4/4 flags with written remediation report.
- **ATTRIB: Supply Chain Vulnerability Attribution** (2024-2025): Extended LLM security research into a 2-phase attribution pipeline that detected defensive workarounds in fuzz-driven OSS patches. Identified and reported 23 P3 workarounds across 10 upstream libraries absent from CVE feeds.
- **LLM Security Research & Tooling** (Fall 2024): Analyzed tools from 15+ papers to automate LLM privacy leakage detection, pen-test command generation, environment-aware vulnerability detection, and AI agent policy enforcement.

Education

Master of Engineering, Cybersecurity (2026) – UNIVERSITY OF MARYLAND, COLLEGE PARK

Bachelor of Science (BS), Mathematics & Bachelor of Engineering, Information Security (2020) – SEOUL WOMEN'S UNIVERSITY

Training & Certifications

- **ISO/IEC 27001:2013 Internal Auditor** – International Register of Certified Auditors (IRCA)
- **Engineer Information Processing** – Human Resources Development Services of Korea
- **Linux Master (Level 2)** – Korea Association of Information & Telecommunication
- **IBM Cybersecurity Fundamentals** – IBM/Coursera
- **Fluent in English & Korean**

Technical Skills

- **SYSTEMS:** Linux, VMware, SGE/SLURM, RAID, SAN, NAS, DAS, NFS, Samba, GlusterFS
- **SECURITY:** Nmap, sqlmap, Hydra, Hashcat, dirb, GDB, pwndbg, Metasploit, CloudGoat, OllyDbg, IDA Pro, Wireshark
- **NETWORKING:** VLAN, VPN, DNS, NAT, 100G InfiniBand
- **CLOUD:** AWS, M365, Azure, Active Directory, WIP/AIP/MDM/MAM, Lambda, EC2, S3, CloudTrail
- **CONTAINERS & DEVOPS:** Docker, Kubernetes, Helm, Singularity, ArgoCD, GitHub
- **MONITORING:** Prometheus, Grafana, Nessus, Burp Suite, Splunk
- **COMPLIANCE:** ISO 27001, 27017, 27018, 27701, GCLP; ISMS-P, NIST 800-53, PCI-DSS, GDPR
- **LANGUAGES:** Python, C, Bash, PowerShell